

Task: KIN - k-inversions

Lab 5. Available memory: 64 MB. Deadline: 30.11.2025, 23:59:59

Description

Let $a_1, \dots, a_n$ be a permutation of numbers from 1 to n . A k -inversion in this permutation is a sequence of indices $i_1, i_2, \dots, i_k$, such that $1 \leq i_1 < i_2 < \dots < i_k \leq n$ and $a_{i_1} > a_{i_2} > \dots > a_{i_k}$. Your task is to determine the number of k -inversions in the given permutation.

Input

The first line of input contains two integers n and k ($1 \leq n \leq 2000$, $2 \leq k \leq 10$)
The second line contains the permutation of numbers $\{1, \dots, n\}$ [cite: 36].

Output

Your program should print the remainder of the division of the number of k -inversions in the given permutation by 10^9 [cite: 38].

Example

Input:

```
4 3
4 3 1 2
```

Output:

```
2
```

Hints

- Initially, propose a solution with time complexity $O(n^2k)$ using dynamic programming[cite: 48].
- Consider how to improve the previous solution to run in $O(nk \log n)$ time using a static segment tree[cite: 49].